OLIVER PINEDA SUÁREZ

Microengineering Graduate
Robotics, Automation, Digital Twins, & Data Solutions

CONTACT

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OliverPSZ

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LANGUAGES

Spanish Native

English C1

French B1

German A1

STRENGTHS

Leadership | Critical Thinking | Problem Solving | Teamwork | Innovation | Adaptability

ACHIEVEMENTS

CENEVAL

National Prize 2019

Awarded for obtaining an outstanding performance in the final exam from the Mexican National Center for Undergraduate Evaluation

Tarleton State University

Top 10 Place (2017 & 2018) in the International CanSat Competition

Nanosatellite competition held by the American Astronautical Society, NASA GSFC, U.S. Naval Research Laboratory and Lockheed Martin

EDUCATION

École Polytechnique Fédérale de Lausanne — **Lausanne, Switzerland**

M.Sc. Microengineering; **GPA: 5.27/6** — **Sep 2022 - Mar 2025**

Minor: Space Technologies
Thesis: Standardized Digital Twin Data Models for Interoperability and Traceability in Space Projects

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Tecnológico de Monterrey — **Mexico City, Mexico**

B.Sc. Mechatronics Eng.; **GPA: 87.27/100** — **Aug 2015 - Dec 2019**

Specialization: Applied Robotics
Thesis: CubeSat Soft Robotic Arm Prototype for Space Debris Removal
Exchange Program: Robotics and Power Electronics at Shibaura Institute of Technology (Tokyo, Japan)

Instituto Politecnico Nacional — **Mexico City, Mexico**

B.Sc. Aerospace Eng.; **GPA: 8.57/10** — **Aug 2015 - Jun 2019**

Specialization: Aircraft Design and Manufacturing
Thesis: Risk Analysis for the Radiographic Testing processes in an Aeronautical Repair Station according to what is established in the 6.1 clause of the ISO 9001:2015 Standard
Exchange Program: Future Space Technologies and Experiments in Space at Samara National Research University (Samara, Russia)

WORK EXPERIENCE

Maxon International — **Sachseln, Switzerland**

Space Systems Digital Twin Engineering Intern — **Aug 2024 - Jan 2025**

• Research and implementation of advanced APIs, interfaces, and semantic models to improve Digital Twin technologies for the space industry.
• Development of guidelines, training materials, and support systems to ensure effective adoption, data validation, and integration of digital twins.

Digital Twin Solutions Engineering Intern — **Feb 2024 - Aug 2024**

• Developed tools for dynamic generation of Industry 4.0 AAS digital twins.
• Built web-based interfaces for data visualization and product traceability.

Nueva Aeronáutica Profesional — **Mexico City, Mexico**

Metrology Laboratory Technical Manager — **Dec 2020 - Aug 2022**

• Development of calibration systems, procedures, uncertainty budgets, and quality assurance for Temperature, Mass, Pressure, Dimensional, Electrical, and Torque.

Aircraft Nondestructive Testing Engineer — **Dec 2019 – Dec 2020**

• Development applications to automate inspection report generation.
• Performed Nondestructive Inspections (VT, ET, MT, PT, RT, UT methods) for various aircraft engines and components.

Software Developer Technical Intern — **Jan 2015 – Dec 2019**

• Development of a Java application for work order management and a Visual Basic (VBA) and Python application to automate financial processes.
• Assist in the development of calibrations and Nondestructive Inspections.

CERTIFICATIONS

SCRUM Fundamentals

AGI STK Level 1

Solidworks Mechanical Design Associate

ASQ Six Sigma Yellow Belt

AFAC Mexico Maintenance Technician Class 1: Airplane, Engines and Airframe

Internal Auditor Based on ISO 19011:2018 Guidelines and ISO 9001:2015 Requirements

SKILLS

Programming Languages & Frameworks:

Python | Java | MATLAB | Simulink | C/C++/C | SQL | Streamlit | QT | Javascript | HTML/CSS | LabVIEW | R | G-code | PLC Programming | BaSyx | ROS | VBA | Dart

Software:

ANSYS | COMET | CATIA | AWS | Siemens TIA Portal | AGI Systems Tool Kit | LaTeX | Orbitron | Eagle | AutoCAD | FeatureCAM | Siemens NX | Solidworks | Qiskit

PROJECTS

EPFL Space Center — **Lausanne, Switzerland**

Software Developer — Sep 2023 – Feb 2024

• Improved the space logistics optimization TCAT tool (developed by eSpace for ESA) to model and simulate lunar and interplanetary space mission scenarios.

Electrical Engineer — Mar 2023 – Mar 2023

• Concurrent Engineering Case Study: Design of the Power Subsystem for the CARINA Lunar Pathfinder Constellation mission proposal.

EPFL Spacecraft Team — **Lausanne, Switzerland**

Lead On-Board Computer (OBC) Engineer — Aug 2023 – Feb 2024

• Developed the CHESS Mission's OBC Preliminary Design Review (PDR) and Design Solution Definition (DSD)
• Led the testing of the Twocan OBC, ensuring its compliance with the specific requirements and objectives of the Twocan In-Orbit Demonstration launched in January 2025 and the upcoming CHESS mission.

Flight Software Engineer — Feb 2023 – Jul 2023

• Semester project focused on the development of flight software (F-Prime component) to interface the X-Band communication system with the On-Board Computer for the upcoming In-Orbit Demonstration.

École Polytechnique Fédérale de Lausanne — **Lausanne, Switzerland**

Software Developer — Dec 2022 – May 2023

• Academic Network Sustainability Simulation: Development of a simulation and infographic based on Mobitool website data to see differences in means of travel in European academic scientific events.

Cuauhtemoc IPN Aeroespacial — **Mexico City, Mexico**

Electronics and Flight Software Engineer — Jul 2017 – Jun 2019

• Development, and testing of CanSat's electronics and flight software.

Lead CDH and Ground Control Engineer — Jul 2016 – Jun 2019

• Development of Ground Station Java-based UI software for telemetry acquisition and telecommand generation.
• Development of digital twins to monitor CanSat's 3D attitude in real time.

Samara National Research University — **Samara, Russia**

Lead & Project Manager — Jun 2019 – Jun 2019

• Space mission proposal (PDR) for a 6U CubeSat using Convolutional Neural Networks to autonomously process and select space debris data.

Shibaura Institute of Technology — **Tokyo, Japan**

Robotics Engineer — Jul 2018 – Jul 2018

• Development of a remotely operated multipurpose robot with real-time object recognition using Convolutional Neural Networks.